Client / Customer : **KIRBYCORP**

Type matériel / Machine type : **SATL 75-169-2**

Data sheet

DATA SHEET ALTERNATEUR SYNCHRONE

GENERATOR TECHNICAL CHARACTERISTICS

D	16/01/2026	L Djoudi		M Shnyparkov	Rated power
C	22/12/2025	L Djoudi		M Shnyparkov	Rated power
B	23/07/2025	L Djoudi		M Shnyparkov	Cooling mode & Machinte type
A	23/06/2025	L Djoudi		M Shnyparkov	Première Edition / First Issue
Rev	Date	Rédacteur/ Written by	Vérificateur/ Checked by	Emetteur/ Issued by	MODIFICATIONS

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1. GENERAL DATA (TO BE CONFIRMED) :

1.1.	Driving machine	Gas turbine
1.2.	Installation	Indoor
1.3.	Climate	Tropical Offshore
1.4.	Site conditions	Marine

2. ELECTRICAL DATA :

2.1.	Rated power	Pn	18000	kW
2.2.	Rated apparent power		22500	kVA
2.3.	Rated voltage	Un	13800	V+/-5%
2.4.	Rated current	In	941	A
2.5.	Rated frequency	f	60	Hz+/-2%
2.6.	Rated power factor		0.8	
2.7.	Rated speed		3600	rpm
2.8.	Overspeed		4320	rpm
2.9.	Number of poles		2	
2.10.	Number of phases		3	
2.11.	Number of H.V. connections		6	
2.12.	Coupling		star	
2.13.	Altitude		<1000	m
2.14.	Ambient temperature mini/maxi		-10/50	°C
2.15.	Cooling fluid type		air	
2.16.	Cooling mode		IC01	
2.17.	Cooling fluid temperature		50	°C
2.18.	Service		S1	
2.19.	Insulation class (stator/rotor)		F/F	
2.20.	Temperature rise class (stator/rotor)		B/B	
2.21.	Norms/standards		IEC	
2.22.	Excitation currents :			
	at no load		290	A
	at full load		644	A
2.23.	Excitation voltages :			
	at no load		47.0	V
	at full load		150	V
2.24.	Efficiencies at rated power factor =0.8			
	at 4/4 load		97.14	%
	at 3/4 load		96.76	%
	at 2/4 load		95.81	%
2.25.	Efficiencies at power factor =1 :			
	at 4/4 load		97.65	%
	at 3/4 load		97.19	%
	at 2/4 load		96.14	%
2.26.	Unsaturated reactances:			
	direct synchronous	Xd	175	%
	direct axis transient	X'd	19.0	%
	direct axis subtransient	X"d	13.4	%
	quadrature axis synchronous	Xq	169	%
	quadrature axis subtransient	X"q	12.6	%
	negative sequence	Xi	13.0	%
	zero sequence	Xo	6.0	%

2.27.	Time constants :			
	direct axis short circuit transient	T'd	0.58	s
	direct axis open circuit transient	T'do	5.4	s
	direct axis short circuit subtransient	T"d	0.054	s
	direct axis open circuit subtransient	T"do	0.076	s
	armature short circuit	Ta	0.112	s
	quadrature axis short circuit subtransient	T"q	0.072	s
	quadrature axis open circuit subtransient	T"qo	0.97	s
2.28.	Saturated reactances :			
	direct axis synchronous	Xds	118	%
	direct axis transient	X'ds	17.1	%
	direct axis subtransient	X"ds	11.4	%
	quadrature axis synchronous	Xqs	113.	%
	quadrature axis subtransient	X"qs	10.7	%
	negative sequence	Xis	11.7	%
	zero sequence	Xos	5.4	%
2.29.	excitation resistance	Raf	0.193	Ω
2.30.	positive sequence armature winding resistance	R1	0.046	Ω
2.31.	negative sequence resistance	R2	0.125	Ω
2.32.	Short circuit ratio	Kcc	0.85	
2.33.	Air gap		25	mm
2.34.	Three phase short circuit current at rated excitation point			
	subtransient component		9.3	Pu
	transient component		6.7	Pu
	permanent		1.9	Pu
	peak value		24.6	Pu
	Constant of inertia	H	1.3	
2.35.	Losses at rated point		530	kW

3. MECHANICAL DATA :

3.1.	Rotor type		Cylindrical-forged	
3.2.	Protection degree		IP 23	
3.3.	Mounting form		IM 1005	
3.4.	Noise level at 1 meter (without acoustic enclosure)		<= 115 +/-2	dB(A)
3.5.	Rotor balancing class		G=2.5 ISO 1940	
3.6.	Weights total		See outline drawing	
3.7.	Generator inertia	MR2	426	kgm2

4. BEARING DATA :

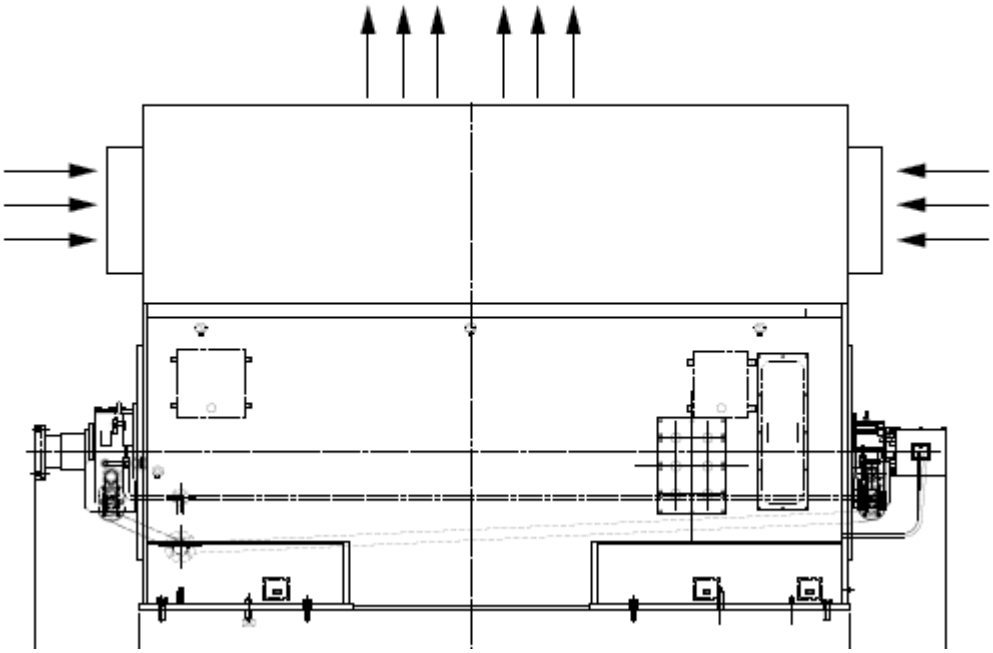
4.1.	Oil pressure drop at bearing inlet	0.8 min	bars
4.2.	Sleeve bearing type	2 lobbes-insulated	
4.3.	Lubrication type	Forced	
4.4.	Bearing losses :		
	Driving End side	21(DE)/18(NDE)	kW
4.5.	Oil flow per bearing:		
4.6.	Driving End side	30	l/min
	Non Drive End side	20	l/min

Lube oil system shall be provided by the customer

4.7.	Oil quality	ISO-VG- 46 or 32	
4.8.	Oil temperature at inlet	50 max.	°C

5. COOLING AIR CIRCUIT DATA :

5.1.	Norms	Standard IC01	
5.2.	Air flow (to be confirmed)	21 to 23	m3/s
5.3.	Inlet filter	Type Munster	



Cold air enter inside the generator from drive end and Non drive end side.
On both side the air shall be within the temperature range indicated in the datasheet (refer to 2.14)
Hot air exhaust can be set in different ways:

- From top middle of the generator (as indicated in the sketch
- From both side right and left (middle of the generator
- From only left side or only right side

The air exhaust can be ducted as discussed during clarification meeting.
A preference has been given from left side only for the air exhaust. This can be set during engineering stage (at an early stage: I.e. during kick off meeting)

5.4. EXCITATION GENERATOR :			
5.4.1. Type		AE 68-6-12	
5.4.2. Number of poles		12	
5.4.3. Ceiling		2	
5.4.4. Response time		<1	s
5.4.5. Protection degree		Integrated to the generator	
5.4.6. Cooling mode		Derivated from main circuit	
5.4.7. Insulation class (stator/rotor)		F/F	
5.4.8. Temperature rise class		B/B	
5.4.9. No load exciter time constant	T'do	0.5	s
5.4.10. Excitation currents :			
at no load		4.5	A
at full load		10.5	A
at ceiling		17.0	A
5.4.11. Excitation voltages :			
at no load		15.3	V
at full load		36.4	V
at ceiling		58.0	V
5.5. PERMANENT MAGNET GENERATOR :		SAP 15.3-6-6	

Permanent magnet generator is feeding the excitation cabinet for the thyristor bridge.
In the excitation cabinet we are foreseen another input, in case of failure of the PMG (as redundancy)
The heating, lighting and control of the excitation cabinet is not fed by the PMG in order to insure availability even when the generator is not rotating.

5.5.1. Number of poles	6	
5.5.2. Ceiling	2	
5.5.3. Response time	<1	s
5.5.4. Cooling mode	Natural	
5.5.5. Insulation class (stator/rotor)	F/F	
5.5.6. Temperature rise class	B/B	
5.5.7. Rated Power	2	kVA
5.5.8. Frequency	180	Hz
5.5.9. No load voltage	225	V
5.5.10. Rated voltage output	215	V
5.5.11. Number of phases	3	
5.6. RECTIFIER ASSEMBLY :		
5.6.1. Type of mounting	Graetz connection	
5.6.2. Rectifier number	12	

6. REGULATOR DATA :

6.1. Type	SARN JE
6.2. Document	Refer to our technical offer

7. ACCESSORIES :

7.1.	Neutral and phase TB boxes	included
7.2.	2 PT100 per phase	included
7.3.	1 PT100 per bearing	Included
7.4.	3 TCs	included
7.5.	3 TPs	included
7.6.	Auxiliary terminal box	included
7.7.	1PT100 at air inlet (both side)	included
7.8.	Neutral earthing resistor	Excluded
7.9.	Vibration probes	Excluded
7.10.	Vibration monitoring	Excluded
7.11.	Lube oil system	excluded (by customer)

Any accessories which are not reported in the datasheet are not included.

8. DOCUMENTS ENCLOSED :

8.1. Curves :

No load saturation and short circuit
Capabilty curve(active/reactive power)
Online short-circuit current
Allowable Negative Sequence Current Versus Time
Allowable Neutral Current Versus Time
Overload capacity

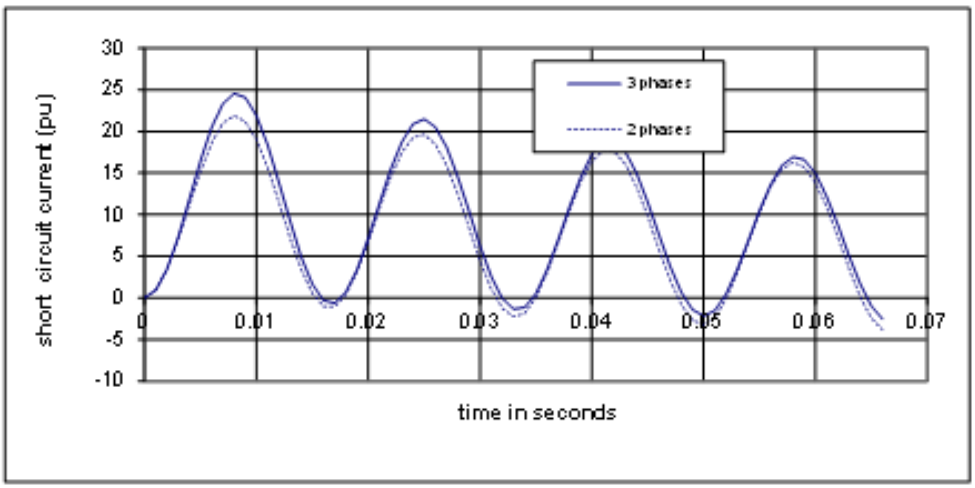
NO LOAD SATURATION AND SHORT CIRCUIT CURVES
COURBE A VIDE ET DROITE DE COURT-CIRCUIT
22500 KVA
UN=13800 VOLTS
IN=941 AMP
NO LOAD SATURATION CURVE
COURBE A VIDE
SHORT CIRCUIT CURVE
DROITE DE COURT-CIRCUIT
IEX/IEX0 (IEX0=289 A)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
1.2
1.3
1.4
1.5
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
3.2
3.4
3.6
3.8
4.0

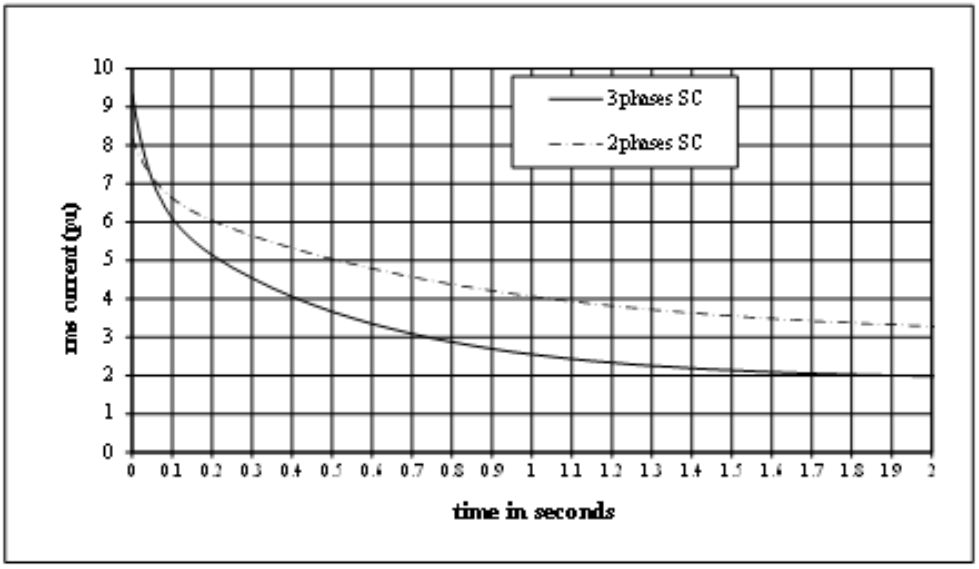
On load Short circuit currents

Data

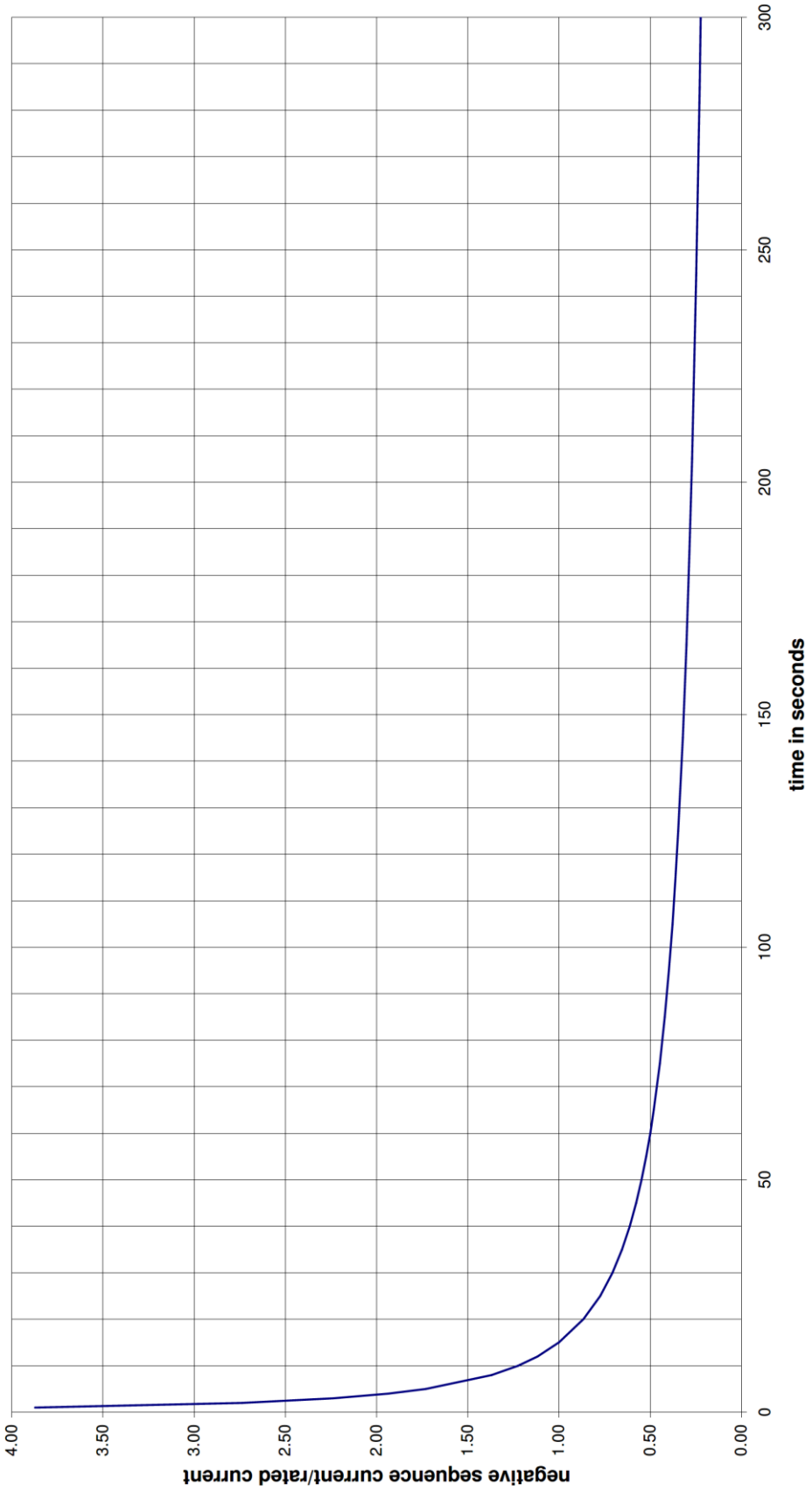
S=	22500	kVa
power factor=	0.8	
frequency=	60	Hz
Voltage=	100%	

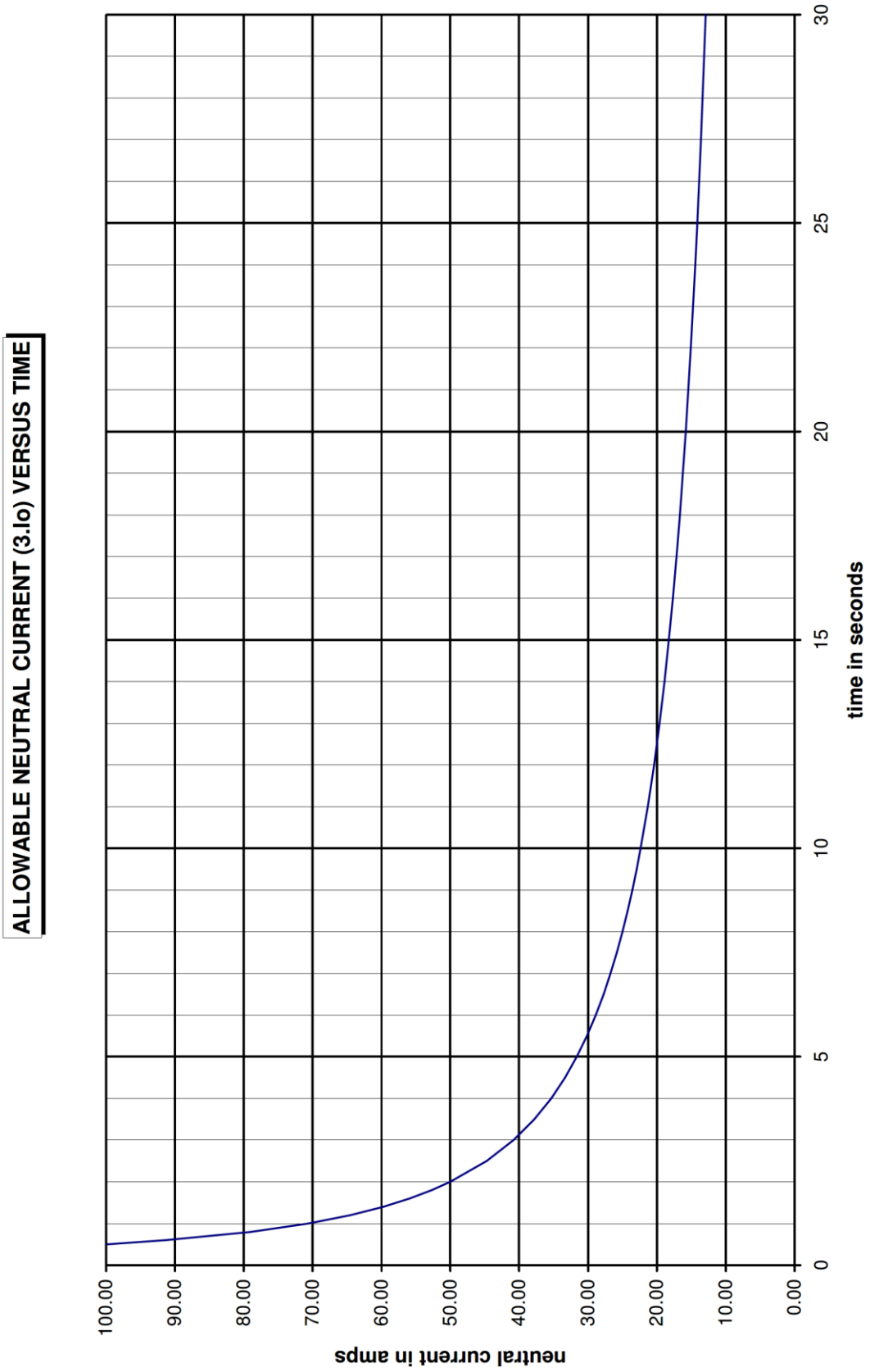
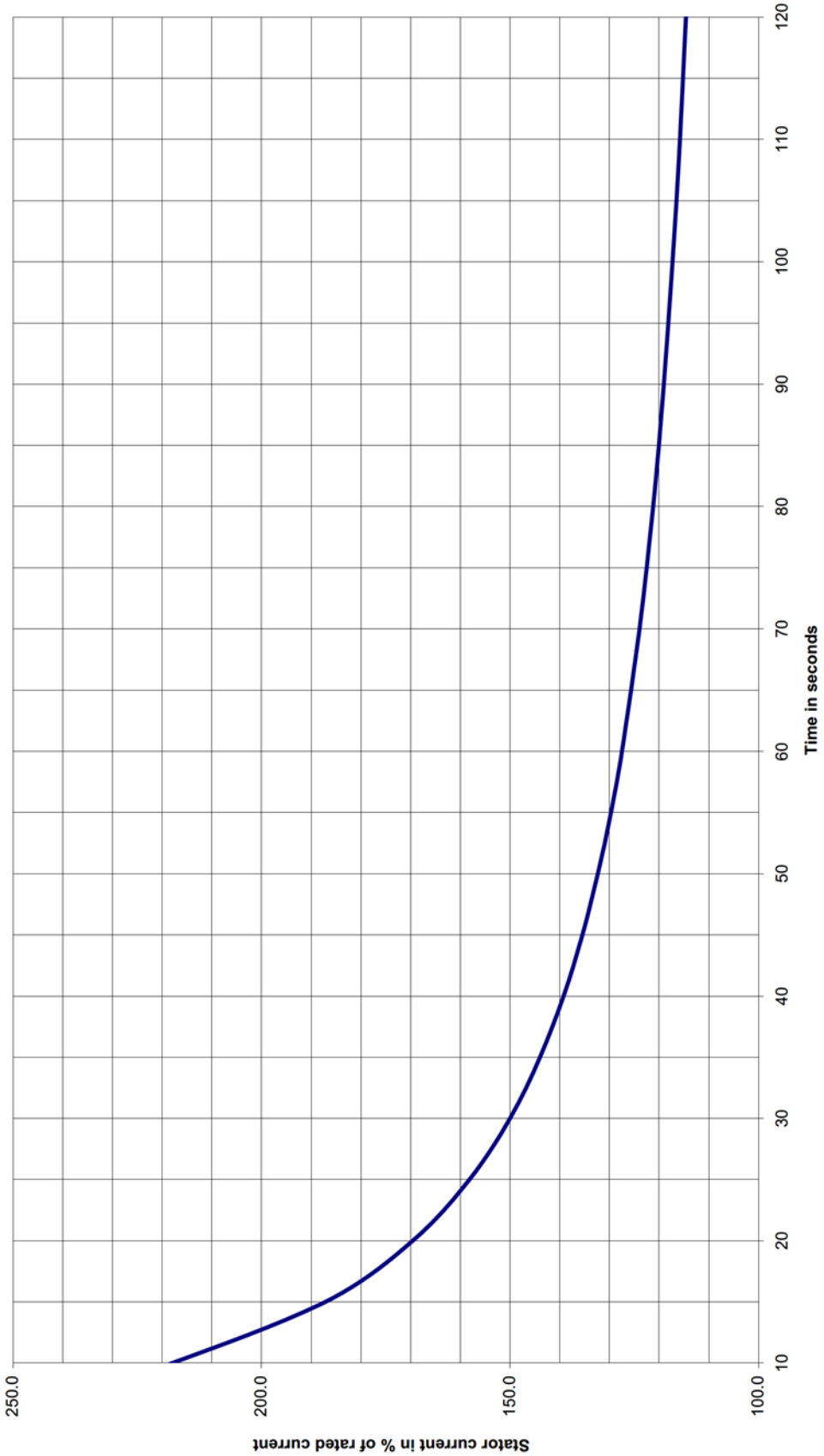
	3 phases		2 phases	
Imax	24.55 pu	23.11 kA	21.88 pu	20.60 kA
subtransient Irms	9.34 pu	8.79 kA	8.19 pu	7.71 kA
transient Irms	6.67 pu	6.28 kA	6.89 pu	6.49 kA
permanent Irms	1.89 pu	1.78 kA	2.99 pu	2.82 kA
asymetric Irms 0.5cycle	14.98 pu	14.11 kA		
asymetric Irms 2 cycles	12.08 pu	11.37 kA		





ALLOWABLE NEGATIVE SEQUENCE CURRENT VERSUS TIME
$(I2 / IN)^2.t = 15$




ALLOWABLE NEUTRAL CURRENT (3.Io) VERSUS TIME
neutral current in amps
100.00
90.00
80.00
70.00
60.00
50.00
40.00
30.00
20.00
10.00
0.00
0
5
10
15
20
25
30
time in seconds

STATOR OVERLOAD CAPACITY